SPACE FACULTY

Satellite Design & Integration

DAY 5

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OBC Exercises

- Send to my ground station all your readings via LoRa separately by typing TEMPOBC, TEMPEPS, TIME, TEMPCOMPARISON and VBAT.

The output should look like:

[Team Name] TEMPEPS = 24.20 deg if you typed “TEMPEPS” or

[Team Name] VBAT = 3.67 V if you typed “VBAT” ...

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OBC Exercises

- Send to my ground station all your readings via LoRa separately by typing TEMPOBC, TEMPEPS, TIME, TEMPCOMPARISON and VBAT.

The output should look like:

[Team Name] TEMPEPS = 24.20 deg if you typed “TEMPEPS” or

[Team Name] VBAT = 3.67 V if you typed “VBAT” ...

- Now, write a new function called *sendAll()* that is capable of sending all the readings in one line. This function has to be activated when typing “ALL”.

The output should look like:

[1m 25s][Team Name] TEMPOBC | TEMPEPS = 24.20 deg | VBAT = 3.67 V

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OBC Exercises

Let's integrate *warnings* into the satellite!

While a satellite remains unmodifiable once launched, it can still be operated to some extent by a ground station.

In addition to ADCS control and operative modes, we can leverage sensor data to monitor the behavior of subsystems.

If the battery level is too low or the temperature is excessively high, we can deactivate certain systems. These decisions are typically made autonomously upon receiving a warning.

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OBC Exercises

Warnings are functions that need to run in the background, constantly monitoring specific conditions. Even if the satellite's status is not actively queried (from the serial monitor), these warnings should appear when such conditions undergo a change.

- Create a warning function that checks the temperatures of both the OBC and the EPS. If either of them detects a temperature higher than 30 degrees Celsius, display the following message in your serial monitor:

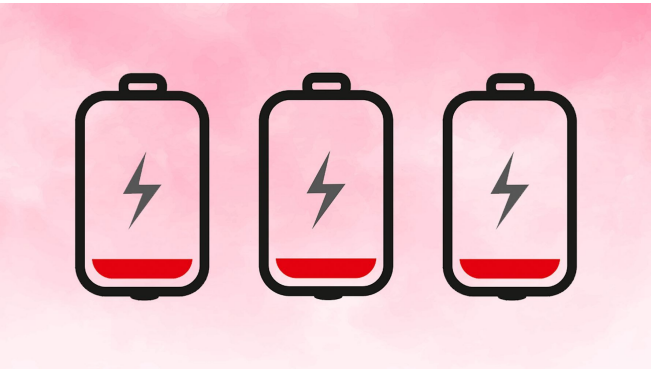
[3m 43s] WARNING! TEMPOBC is too high: 35 deg! or
[3m 43s] WARNING! TEMPEPS is too high: 35 deg!

We can use the rework to test it out!

OBC Exercises

- Create a warning function that checks the voltage of the battery on the EPS. If the value is found to be lower than 3.6 V, display the following message in your serial monitor:

[3m 43s] WARNING! VBAT is too low: 3.5 V!



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OBC Exercises

Add to the code the possibility of typing “STATUS” to get an output that looks like this:

- [3m 43s] NOMINAL — if no warnings are found
- [3m 43s] WARNING! TEMPxx is too high: 35 deg! — if temperature is too high
- [3m 43s] WARNING! VBAT is too low: 3.5 V! — if battery is too low

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OBC Exercises

Add to the code the possibility of typing “STATUS” to get an output that looks like this:

- [3m 43s] NOMINAL — if no warnings are found
- [3m 43s] WARNING! TEMPxx is too high: 35 deg! — if temperature is too high
- [3m 43s] WARNING! VBAT is too low: 3.5 V! — if battery is too low

Now, in the *sendAll()* function include the STATUS of the satellite in two lines.

[2m 23s] [Team Name] TEMPOBC | TEMPEPS = 24.20 deg | VBAT = 3.67 V |
[2m 23s] [Team Name] STATUS = one_of_the_cases_above

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OBC Exercises

- Implement a code to monitor the STATUS so that, even when we don't explicitly request 'ALL' or 'STATUS,' if the satellite detects a warning, it will send a message to the ground station via LoRa.

[6m 15s] [Team Name] STATUS = WARNING! VBAT is too low: 3.5 V!

When you are done **SAVE YOUR SKETCH AS OBC_DAY5**!

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Solar Panels

A solar panel, or photovoltaic panel, is a device that converts sunlight into electricity.

The basic unit of a solar panel is the solar cell which is made of semiconductor materials.

It is known as “photovoltaic effect” the process through which the sunlight generates electric current.

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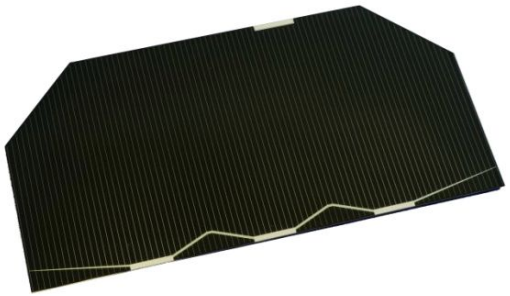
Space Grade Solar Panels

Conventional solar cells used for civilian applications typically utilize a double-junction design. In contrast, triple-junction solar cells, preferred in space missions, prioritize heightened efficiency over concerns of cost and weight.

Engineered to effectively capture a wide spectrum of solar radiation, these cells excel in the challenging conditions of space with limited sunlight. Triple-junction cells significantly enhance energy absorption and power generation.



Their durability in harsh space environments, including exposure to radiation and extreme temperatures, positions them as an optimal choice for powering satellites, space probes, and scientific instruments.

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Space Grade Solar Panels

If we consider a double row of solar cells on each face of a CubeSat, the number of cells may be distributed as follows:

- **1U CubeSat**: A 1U CubeSat with a double row of solar cells might have around 6 to 12 solar cells.
- **2U CubeSat**: For a 2U CubeSat, a double row could result in approximately 12 to 24 solar cells.
- **3U CubeSat**: A double row on a 3U CubeSat might feature around 18 to 36 solar cells.
- **6U CubeSat**: On a 6U CubeSat, a double row configuration could include approximately 36 to 60 solar cells.
- **12U CubeSat**: For a 12U CubeSat, with a double row, you might find around 60 to 120 solar cells (doployables).

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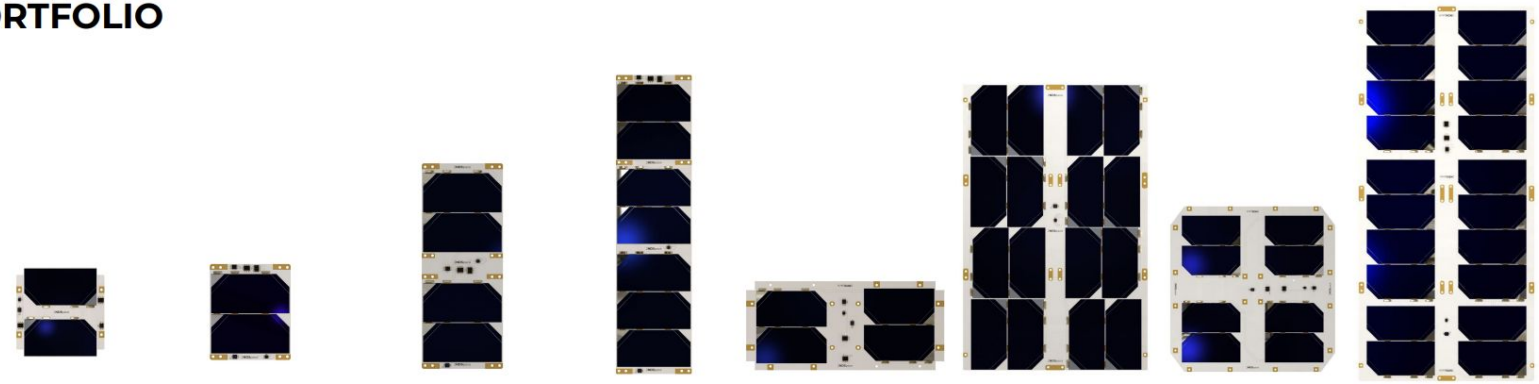
PRODUCT **PORTFOLIO**



	CORE-01Z	CORE-01	CORE-02	CORE-03	CORE-06Z	CORE-06	CORE-12Z	CORE-16
Peak Power [W]	2.5	5	5	8,75	5	20	10	25
Suggested for	1U/2U/3U	2U/3U	2U/3U	3U/6U	6U	6U/12U	12U/16U	16U
Form Factor	1x1	1x1	1x2	1x3	2x1	2x3	2x2	2x4
Weight [g]	38	35	75	118	91	289	195	387

QUALIFICATION AND **ACCEPTANCE TEST**

	Functional/AIT	Electrical Test/Flash	Vibration test	Mechanical Shock	TVAC Test
Qualification Test	✓	✓	NASA GEVS: GSFC-STD-7000A ESA ECSS-E-ST-10-03C	NASA GEVS: GSFC-STD-7000A ESA ECSS-E-ST-10-03C	NASA GEVS: GSFC-STD-7000A ESA ECSS-E-ST-10-03C
Acceptance Test	✓	✓			

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Space Grade Solar Panels

Let’s have a look to a real 1U solar panel!

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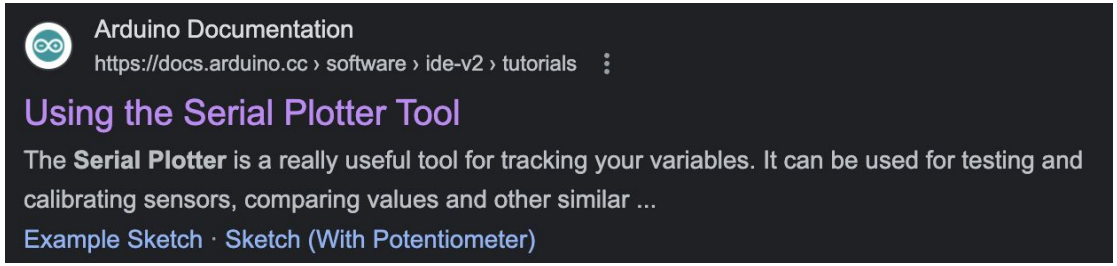
Back to the OBC!

To ensure consistency in sensor readings, let's explore the Serial Plotter in the Arduino IDE. **There is no need for LoRa. Let's test this on a new sketch.**

Let's have a look at the Serial Plotter on the Arduino IDE. Refer on its documentation which is accessible online on the Arduino website.



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Back to the OBC!

To ensure consistency in sensor readings, let's explore the Serial Plotter in the Arduino IDE. **There is no need for LoRa. Let's test this on a new sketch.**

Let's have a look at the Serial Plotter on the Arduino IDE. Refer on its documentation which is accessible online on the Arduino website.

- Utilize the serial plotter to graphically represent all readings, including TEMPOBC, TEMPEPS, and VBAT.
- Remember that the reading must be continuous to correctly plot the values.
- Find a way to calculate the variability of the sensor readings within a 5 seconds as sample time.

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OBC Exercises - Camera Challenge

The satellite has a serial camera module called **Adafruit VC0706**.

By referring to its online documentation:

- Create a **new sketch**.
- Refer to the OBC schematics to understand the camera connection to Arduino.
- Create a code that allows you to take a picture and show it in its “text” form. We will see how to convert it to JPG!


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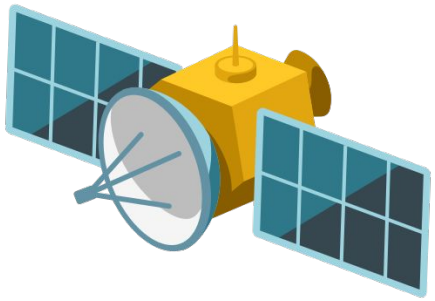
Satellite Design & Integration

END of DAY 5



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